

Residence Time Calculation

"SSP": Stable States picture is the main highlight; but we'll talk about other schemes as well.

Impey, McDowell: 3 scientists

① IMM scheme (1983)

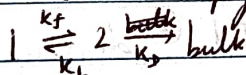
↑
Madden

define $p_\alpha(r, r+t; t^*)$: water molecule α lies in the 1st hydration shell of ref. molecule x at $t \in (r, r+t)$ w/o crossing it for time τ^*

Then using this definition we can use:

$$C_{IMM}(t, t^*) = \frac{\langle p_\alpha(r, r+t; t^*) \rangle_{\alpha, r}}{\langle p_\alpha(r, r; t^*) \rangle}$$

② Kinetic Model (1 & 2 imply 1st and 2nd hydration shells)



$$\frac{dc_1}{dt} = -k_f c_1 + k_b c_2 \quad ; \quad \frac{dc_2}{dt} = k_f c_1 - k_b c_2 - k_d c_2$$

(coupled)

We're a system of linear ODEs. Either we can take a 2nd order derivative of one of these or we can find the e-values of the matrix used to represent the ODE system.

for our system:

$$\begin{pmatrix} dc_1/dt \\ dc_2/dt \end{pmatrix} = \overset{A}{\begin{bmatrix} -k_f & k_b \\ k_f & -(k_b + k_d) \end{bmatrix}} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

The e-values will give the decay rates of the concentrations

$$\det(A - \lambda I) = 0 \Rightarrow (k_f + \lambda)((k_b + k_d) + \lambda) - k_f k_b = 0$$

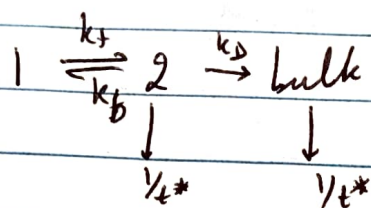
$$k_f k_d + \lambda(k_f + k_b + k_d) + \lambda^2 = 0$$

$$\lambda = \frac{-(k_f + k_b + k_d) \pm \sqrt{(k_f + k_b + k_d)^2 - 4k_f k_d}}{2}$$

$$\text{Relax}^n \text{ Time } \tau_{\text{decay}} = -\frac{1}{\lambda} = -\frac{1}{\text{decay}}$$

$$\therefore \tau_f = \frac{1}{\lambda} = \frac{1}{\frac{-(k_f + k_b + k_d) \mp \sqrt{(k_f + k_b + k_d)^2 - 4k_f k_d}}{2}}$$

③ Addition to Kinetic Model (to make it similar to IMM's t^*).



: This $1/t^*$ is the relaxⁿ rate

Now, for 2, we have an additional rate, so the governing eqⁿs for $\tau'_\pm$ are:

$$\frac{1}{\tau'_\pm} = \frac{1}{2} (k_f + k_b + k_D + \frac{1}{t^*} \mp \Delta') \quad ; \quad \Delta' = \sqrt{(k_f + k_b + k_D + \frac{1}{t^*})^2 - 4k_f(k_D + \frac{1}{t^*})}$$

$$\tau^{res} = \tau'_+ \approx \frac{1}{k_f} [1 + k_b t^* + k_b (k_f - k_D) t^{*2} + O(t^{*3})]$$

(why + only? see ref 27 p16)

Now, a plot of τ^{res} is generated @ different t^* values & then from $\tau^{res}(t^*)$, k_f is obtained by curve fitting. (using IMM, 1983 scheme)

$$MR7 = 1/k_f$$

④ Reactive Flux approach (from David Chandler's ^{Intro to} Modern Statistical Mechanics)

$$\frac{1}{k^{RF}(t)} = \left[\frac{\langle \dot{p}(0) (1 - p(t)) \rangle}{\langle p \rangle} \right]^{-1}$$

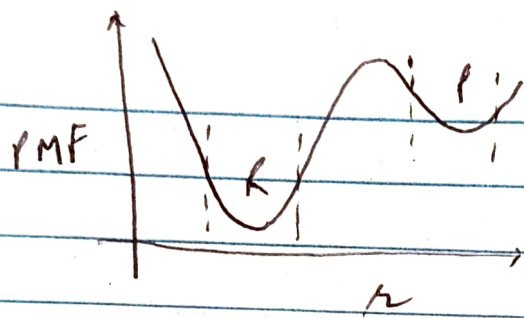
$p(t) = p_A(t, t; 0)$; $\dot{p}(t)$ being the t-derivative of $p(t)$

~~1/k^{RF}(t)~~

$\frac{1}{k^{RF}(t)}$: Supposed to reach a plateau value @ longer times (t).

In long time limits: $\frac{1}{k^{RF}(t)} \rightarrow \frac{1}{k_f}$ but; we still have some effect of k_b ; and that makes it not-so-useful.

⑤ SSP: Stable State Picture



So, in SS approach, we define stable reactant and stable product regions.

And for this approach:

$$1 - \langle I_R(0) I_P(t) \rangle = \exp(-t \tau_{ssr}^{us})$$

And τ_{ssr}^{us} is not very sensitive to the location of the boundaries of stable R and stable P.